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Machine Learning Approaches for Near-Surface Velocity Model Building

H. Alqatari, E. Turkoglu, D. Colombo, and E. Sandoval Curiel, EXPEC Advanced Research Center – Saudi Aramco

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Abstract

Near-surface complexities pose major challenges for land seismic data processing and interpretation. Accurate modeling of the near-surface velocity is key to compensate the effects of these shallow heterogeneities, which can distort seismic signals and reduce image quality. Uphole data are of great importance in improving the estimation of the near-surface velocity model as they provide direct vertical travel time measurements of the seismic wave traveling through the weathered layers. Near-surface attributes, namely velocity, depths and thicknesses, can be characterized by the interpretation of the uphole travel time data. In this study, we develop an automated velocity model-building method of uphole data based on neural networks to address the need for consistency and efficiency. The goal is standardizing and automating what is otherwise a laborious and error-prone interpretation workflow that often relies on subjective analyst input. The neural network model, trained with synthetically generated uphole data, is able to construct velocity profiles from uphole travel time surveys by learning from representative patterns in the input data. We demonstrate the effectiveness of the trained model in producing robust velocity predictions using synthetic as well as real uphole datasets, highlighting its potential for improving near-surface velocity characterization and supporting seismic processing in complex environments.

Introduction

Near-surface modelling involves describing the Earth's topmost layers, often termed the weathering zone, which are characterized by loose, unconsolidated sediments with low velocity, variable thickness, and high absorption of the seismic energy. It is particularly challenging in arid regions in which sand dunes, karsts, wadis, and sabkhas dominate the surface, adding another level of complexity to the processing of the seismic waves by significantly altering the propagation. These surface conditions often lead to strong spatial variability in near-surface properties over short distances, making it difficult to apply uniform static corrections. Seismic wavefields traveling through a complex near-surface area exhibit time delays, distortion of their amplitudes and may possibly create false time structures, biasing the interpretation of stratigraphic traps or low-relief structures. Travel-time corrections (Cox, 1999) of the seismic trace must be applied to compensate these near-surface effects, which can otherwise compromise the reliability of the entire imaging workflow. Therefore, the accurate representation of the near-surface velocity model is essential to obtain a reliable final seismic image and reduce interpretation risk.

Much effort has been made to improve the estimation of the near-surface velocity field. Ray-based refraction tomography (Hampson and Russell, 1984; Stefani, 1995), waveform tomography (Tarantola, 1984) and multiphysics joint inversion (Colombo, 2016) are all indirect methods used to construct the shallow velocity model. While these approaches have seen widespread use, their performance is highly dependent on data quality, spatial coverage and the presence of strong refractors. This limitation highlights the need to supplement these techniques with direct velocity measurements obtained from uphole surveys, which provide reliable reference data for velocity calibration.

Uphole survey is a shallow seismic method, employed by lowering an array of explosive sources inside a borehole and by deploying an array of receivers at the surface (Figure 1). Information about the near-surface interval velocities, depths, and thicknesses is interpreted based on the uphole arrival times of the seismic waves generated at the source and transmitted through the interfaces between geological layers of differing velocities. These measurements provide vertically constrained velocity information with high resolution in the shallow subsurface, making them particularly useful for refining near-surface models and improving static corrections. When integrated with other geophysical methods, upholes can serve as reliable calibration points, particularly in geologically complex terrain where indirect methods may struggle to resolve vertical velocity structure.

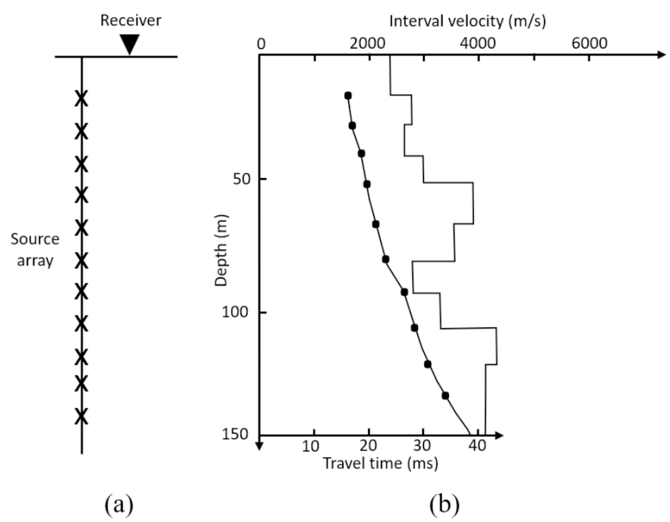

Figure 1—A schematic diagram showing (a) uphole survey layout with a single receiver and a source array and (b) the recorded uphole travel time data combined with the interpreted interval velocity (modified after Marsden, 1993).

The interpretation of uphole travel time data to construct a velocity profile includes manually analyzing the plotted uphole time-depth measurements, line segments, and breaking points to infer the depth of each identified geological layer and utilizing the slope reciprocals to form a velocity-depth model (Boulding, 1996). Repeating the process for each individual uphole dataset is a time-demanding task, especially when processing large fields containing over 2000 uphole locations. Moreover, it is a subjective procedure that varies from one interpreter to another, which, combined with the amount of noise contaminating the uphole data, can lead to spatially inconsistent velocity determinations and limit confidence in the results. Inconsistent picks or interpreter bias may introduce further inaccuracies that propagate into the final model. To overcome the limitations introduced by the interpretation process, we design a robust machine learning neural network model that automatically produces interval velocity models with a generalization capability to unseen data and different levels of noise.

Artificial Neural Networks

Artificial neural networks are a type of machine learning algorithms designed to simulate the human brain in terms of structure and function. Following their early introduction as a simple neuron model by McCulloch and Pitts (1983), artificial neural networks have undergone significant development and now serve as powerful computational systems applicable to a wide range of scientific fields and real-world problems. They are configured to explore nonlinear relationships and identify hidden patterns within representative datasets to produce a desirable outcome, often in the form of classification, regression, or prediction. Their adaptability makes them particularly useful for modeling complex systems where explicit analytical solutions are difficult or impractical to obtain.

Artificial neural networks are composed of interconnected processing units, also known as nodes or neurons, and are organized in layers: an input layer, one or more hidden layers, and an output layer. The output of each neuron within one layer is passed as the input to the neurons in the subsequent layer (Figure 2a). Emulating the synaptic connections between brain cells, the artificial neurons are connected through modifiable connection weights, which modulate the influence of each input on the neuron's output. In addition to the weighted inputs, each artificial neuron includes a bias term and a transformation function, collectively allowing the network to approximate highly nonlinear mappings between inputs and outputs.

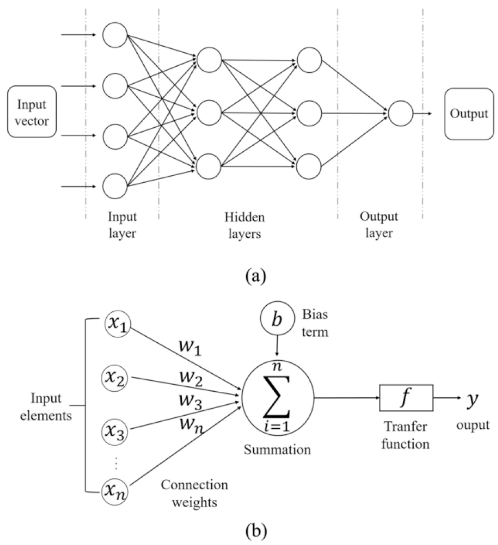

Figure 2—(a) Feedforward neural network model architecture with two hidden layers and (b) an artificial neuron model.

Assuming a fully-connected neural network, once the raw input data is passed through the input layer to the first hidden layer, each neuron in that layer processes the information by summing the weighted inputs, adding a bias, and applying a nonlinear transformation function to produce a scalar output (Figure 2b). This output is then propagated forward as input to the next layer. The process is repeated through all layers until the final output is generated. The choice of activation function plays a critical role in determining the behavior of the network, enabling it to learn complex, non-linear decision boundaries.

For a neural network with n number of input elements, the output signal y of an artificial neuron can be formulated as:

$$y = f\left(\sum_{i=1}^{n} \boldsymbol{w}_i \boldsymbol{x}_i + b\right), \tag{1}$$

where $\boldsymbol{x}_i$ denotes the input vector, $\boldsymbol{w}_i$ denotes the associated connection weight vector, b is the bias term and f represents the transfer function of choice, also known as the activation function.

Feedforward neural networks, often associated with a supervised learning style, include transferring the information in a forward direction: from the input layer to the output layer. The training of a supervised neural network includes randomly initializing all weights and biases, followed by feeding it with a set of labeled inputs and desired outputs (i.e., training dataset). The network processes the data as described earlier to produce its output results, which are then compared against the desired outputs. The mismatch between the actual and predicted values is calculated through an error or loss function, and the resulting error is propagated back through the network using a backpropagation algorithm to adjust the model parameters. This process is repeated iteratively until the network reaches an acceptable level of accuracy or convergence.

The trained neural network model is then used to make predictions about unseen data. Ideally, the training dataset is representative enough so that the trained model is neither underfitting or overfitting; in other words, the model is able to capture the complex patterns in the training dataset while maintaining the ability to handle new data and make accurate predictions (i.e., generalization). Regularization techniques to the neural network learning algorithms can aid in improving their performance and obtaining an optimal fitting. For a comparative analysis on the commonly used regularization techniques, we refer to Nusrat and Jang (2018).

Method

Given that machine learning algorithms, including neural networks, require large and diverse training datasets to make robust and generalizable predictions, and considering the limited availability of extensive field uphole survey data, we opted for a training dataset based on synthetically generated uphole surveys. This approach allows the model to learn a broader range of velocity behaviors that may not be fully captured by field data alone, especially in geologically complex areas where uphole measurements are sparse or unevenly distributed.

Uphole data modelling involved the creation of 40,000 synthetic velocity-depth profiles, each generated by sampling from realistic statistical distributions of layer velocities and thicknesses. Using these values, the corresponding uphole travel times were computed analytically. The simulation was performed down to a depth of 1000 meters, capturing the entire range of shallow geophysical interest. To make the dataset more representative of real-world scenarios, a 5% random noise was added to the simulated uphole time-depth information. This perturbation simulates field acquisition errors and serves as a form of regularization during training, helping to prevent overfitting and improve the model's ability to generalize to unseen data (Sietsma & Dow, 1991; Reed & Marks, 1999).

In this framework, the noisy uphole travel time samples were used as input to the neural network, while the corresponding synthetic interval velocity profiles served as target outputs. Prior to model training, a data normalization step was applied to both inputs and targets. This ensures that all variables are brought to a consistent scale, improving the efficiency and numerical stability of the training process. Normalization also aids in achieving faster convergence and avoids the pitfalls of local minima (Shanker & Hung, 1996; Sola & Sevilla, 1997). Data normalization can take many forms, one of which is data rescaling, also known as the min-max normalization, the method of choice in this study, which linearly transforms the inputs and the targets in the range [-1,1]. It is formulated as follows:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}, \tag{2}$$

Following normalization, the full dataset was randomly divided into three subsets: 70% for training, 15% for validation, and 15% for testing. Each training input consisted of a one-dimensional travel time vector corresponding to a single uphole profile, sampled at uniform 2-meter intervals. The target vector was composed of interval velocities, sampled at variable intervals, finer at shallower depths and coarser at deeper sections.

The neural network model was constructed with a shallow architecture, defined by a single hidden layer. The hidden layer contained 30 neurons and employed a tan-sigmoid activation function to capture nonlinear relationships in the data. The output layer used a linear activation function suitable for continuous regression tasks such as velocity prediction (Figure 3). For training, we adopted the Levenberg-Marquardt back-propagation algorithm, a well-established minimization optimizer known for its fast convergence and stability. This made it an appropriate choice for our function-fitting application involving high-dimensional velocity data (Hagan & Menhaj, 1994).

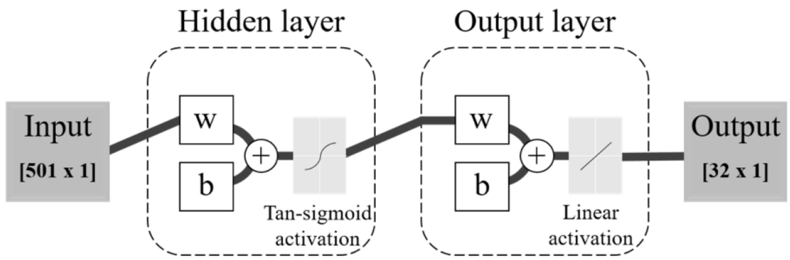

Figure 3—The employed neural network model architecture.

Results and Discussion

Performance assessment of the trained neural network model was carried out by providing a measure of similarity between the actual and predicted variables via the correlation coefficient R, which quantifies the linear relationship between two datasets. Figure 4 illustrates this relationship through scatter plots, indicating the correlation on the training, testing and validation subsets, as well as on all the data combined. R holds a value ranging between −1 and 1; a perfect correlation is at 1, indicating ideal agreement between predicted and actual values. All subsets indicate strongly correlated pairs of actual and predicted velocity values, suggesting that the network has effectively captured the underlying mapping between input travel times and output velocity structures. The RMSE was calculated to be 203 m/s, which reflects a low overall prediction error, further validating the network's capability.

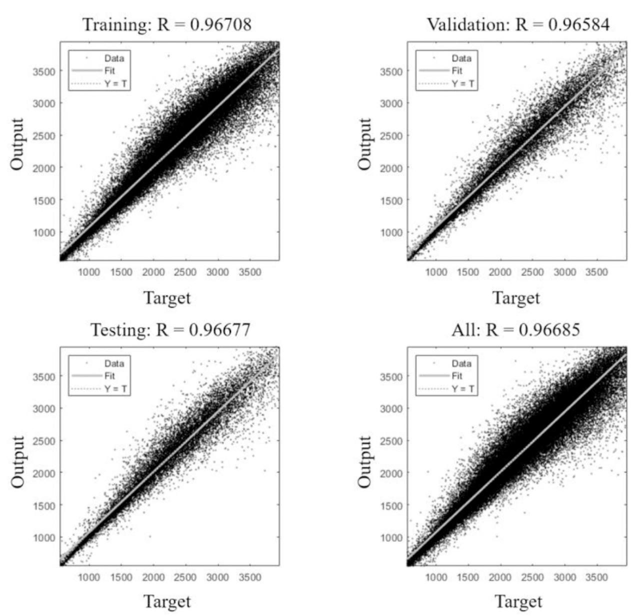

Figure 4—Correlation coefficient R calculated on the training, testing, validation and all the data sets.

The performance of the trained neural network was evaluated by computing the cross-correlation coefficient (R), as well as the root-mean-square error, formulated as:

$$R = \frac{\sum_{i=1}^{N}(x_i - x_m)(y_i - y_m)}{\sum_{i=1}^{N}(x_i - x_m)^2 \sum_{i=1}^{N}(y_i - y_m)^2}, \tag{3}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - x_i)^2}{N}}\,. \tag{4}$$

where x_i represents the actual value (velocity), y_i represents the predicted value, x_m and y_m represent the mean of the actual and predicted datasets, respectively, and N denotes the total number of data points.

The trained model was tested on synthetic as well as on uphole datasets to assess both controlled and real-world performance. Figure 5a illustrates its application on unseen synthetic uphole travel time data (top figure) and the predicted velocity, marked as dashed grey line, in the bottom figure, combined with the true interval velocity, plotted as a solid line. It can be observed that the trained network model is able to produce a velocity model that matches well with the actual profile, indicating successful learning and generalization. Furthermore, figure 5b demonstrates the effect of using noisy uphole travel time data on the velocity response obtained with the trained network model. The input data in this example is the same dataset utilized in figure 5a with the addition of noise, simulating more realistic acquisition conditions. The network model is still able to produce a comparable velocity to the true one, demonstrating its ability to robustly handle data with various levels of noise without significant degradation in output quality.

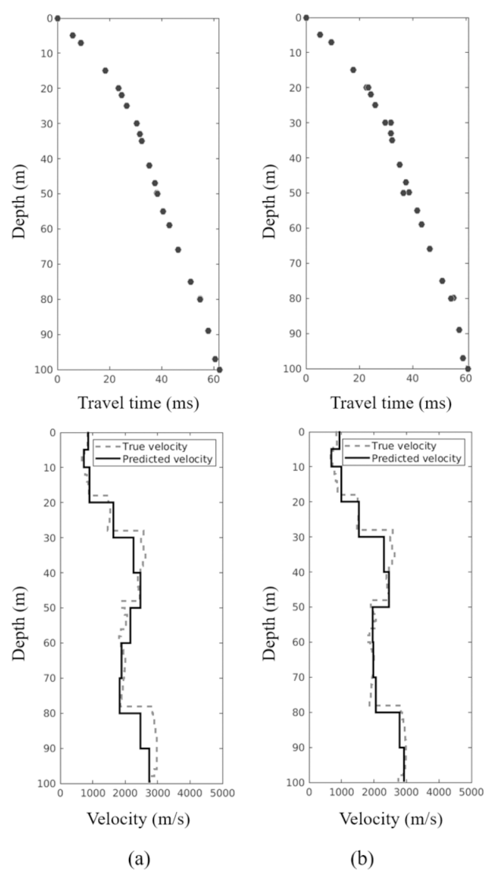

Figure 5—Synthetic application of the trained neural network model, demonstrating the result of mapping (a) uphole travel time data (top) to interval velocity (bottom, dashed grey line), and (b) noisy uphole travel time data (top) to interval velocity (bottom).

Following an extensive testing of the trained model using synthetic data, we applied it to field data to verify its performance. Figure 6 illustrates the result of predicting interval velocity models from 242 uphole field data. The results are compared against a high-resolution near-surface velocity depth slice obtained through Laplace-Fourier full waveform inversion (Colombo et al., 2021), which serves as a reliable reference model. The automatically generated uphole velocity profiles using the trained neural network model show consistent spatial responses and are in agreement with the LF-FWI model, suggesting that the model preserves essential lateral velocity variations. This is more evident in the western part (left portion of the model), where the interpreted upholes indicate higher near-surface velocities consistent with the LF-FWI model, supporting the model's reliability and interpretability.

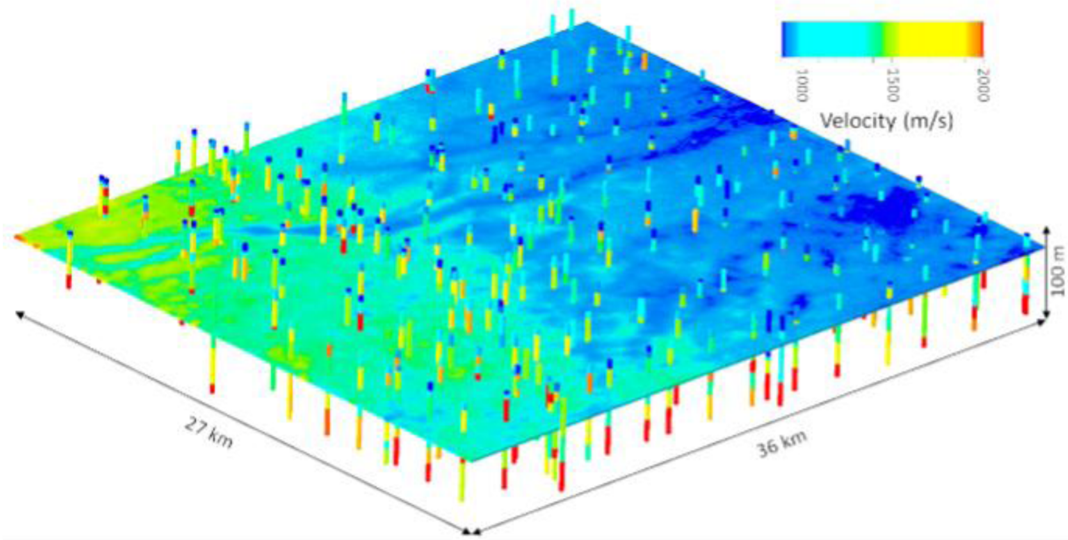

Figure 6—Near-surface velocity model obtained using 1.5D Laplace-Fourier full waveform inversion (LF-FWI) compared to the trained neural network model predicted velocity profiles shown as 242 vertical functions.

Conclusion

We used a physics-based supervised learning method and trained a feedforward neural network model to automatically predict interval velocity profiles from uphole data, aiming to address the challenges of manual interpretation and limited scalability. This approach leverages the physical relationships between travel times and subsurface velocities while benefiting from the learning capabilities of neural networks. Applications of the developed model succeed in producing accurate and robust results in both synthetic and field data, demonstrating its practical viability across different acquisition conditions, including those with varying geological complexity. Noise integration, combined with careful sampling of the training inputs, improves the generalization capabilities of the trained network by exposing it to a wide range of input scenarios during training, thereby reducing the risk of overfitting. The consistently stable predictions, despite varying levels of noise, further validate the effectiveness and reliability of the model in realistic settings, where data quality can be uncertain or variable. Prediction results from the developed network model are well comparable with a high-resolution LF-FWI velocity model, reinforcing the model's ability to capture lateral velocity variations and produce geologically meaningful outputs. This agreement supports the potential of the method to supplement or accelerate high-resolution modeling efforts in operational workflows. The developed method presents a new potential for improved efficiency in velocity model building by enabling the re-utilization of vast amounts of archived uphole data for enhanced near-surface characterization in seismic imaging workflows, reducing both the time and manual effort typically required in conventional approaches.

Nomenclature

y: Output signal from a neuron.
$\boldsymbol{x}_i$: Input feature to a neuron.
$\boldsymbol{w}_i$: Weight associated with input $\boldsymbol{x}_i$.
b: Bias term.
f: Activation function.
$RMSE$: Root-Mean-Square Error (m/s).
R: Correlation coefficient.

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